

ECS401: Unit 6 Lab Progress Test

FEEDBACK: Answers and Explanations

Work hard and you will eventually be able to program. You should have written dozens and dozens of small programs by now! If you got any wrong you urgently need to do something to improve your understanding.

If you got any wrong you urgently need to do something to improve your understanding.

1. Draw the dry run table and give the text that is printed to the screen after the following fragment of code is executed.

```
int z = 1;           \\ L1
while(z <= 5)        \\ L2
{
    z = z + 1;       \\ L3
}
System.out.println(z); \\ L4
```

DRY RUN TABLE:

Line	z	z <= 5 ?	print
L1	1		
L2		1 <= 5 ? true	
L3	2		
L2		2 <= 5 ? true	
L3	3		
L2		3 <= 5 ? true	
L3	4		
L2		4 <= 5 ? true	
L3	5		
L2		5 <= 5 ? true	
L3	6		
L2		6 <= 5 ? false	
L4			6

CORRECT ANSWER:

6

EXPLANATION:

The loop says that while z is less than or equal to 5 it should do the body again. z starts from 1 (so it does enter the body of the loop the first time) and the body adds

one to `z` each time. Eventually `z` will have been increased one at a time until it is 5. At that point the test is still `true` so the body is entered again. That increases `z` to 6 and the test is checked again. This time it is `false` so the loop exits and the value of `z` (ie 6) is printed.

2. Draw the dry run table and give the text that is printed to the screen after the following fragment of code is executed.

```
int z = 1;           \ \ L1
while(false)         \ \ L2
{
    z = z + 1;       \ \ L3
}
System.out.println(z); \ \ L4
```

Line	z	false ?	print
L1	1		
L2		false ? false	
L4			1

CORRECT ANSWER:

1

EXPLANATION:

`z` is set to 1. The loop test is `false`. The body of a while loop is only entered if the test evaluates to `true`, so the loop is not entered at all. The print statement is executed immediately leading to that initial value of 1 that is still in `z` being printed.

3. Draw the dry run table and give the text that is printed to the screen after the following fragment of code is executed.

```
int z = 1;           \\ L1
while(true)          \\ L2
{
    z = z + 1;        \\ L3
}
System.out.println(z); \\ L4
```

Line	z	true ?	print
L1	1		
L2		true ? true	
L3	2		
L2		true ? true	
L3	3		
L2		[FOREVER]	

CORRECT ANSWER:

EXPLANATION:

The print statement is after the loop and it will only be executed when the loop finishes. However the loop continues while its test is `true` and will only stop when the test becomes `false`. However nothing in the loop can make the test become `false`. It will never do so, meaning the loop will never exit. The print statement will never be executed and so nothing will appear on the screen.

4. Draw the dry run table and give the text that is printed to the screen after the following fragment of code is executed.

```
int z = 1;           // L1
int sum = 0;          // L2
while(sum < 12)        // L3
{
    sum = sum + z;      // L4
    z = z + 1;          // L5
}
System.out.println(sum); // L6
```

Line	z	sum	sum < 12 ?	print
L1	1			
L2		0		
L3			0 < 12 ? true	
L4		1		
L5	2			
L3			1 < 12 ? true	
L4		3		
L5	3			
L3			3 < 12 ? true	
L4		6		
L5	4			
L3			6 < 12 ? true	
L4		10		
L5	5			
L3			10 < 12 ? true	
L4		15		
L5			6	
L3			15 < 12 ? false	
L6				15

CORRECT ANSWER:

15

EXPLANATION:

The loop keeps going while the sum has a value less than 12. Each time it adds the new value of z on to sum. z starts at 1 and just increases so first 1 is added to sum, then 2 is added to sum, then 3 is added to sum and so on. sum therefore starts at 0 then becomes 1 the first time round the loop, then becomes 3 (as 2 is added), then becomes 6 (z is now 3 which is added to the 3 in sum), then 10 (as 4 is now added), then 15 (as 5 is added). Up to this point the value of sum has always been less than 12 so the loop has continued, but now sum is not less than 12 so the loop does not execute again and the print statement is executed instead, printing the 15 that is now in sum.

5. Draw the dry run table and give the text that is printed to the screen after the following fragment of code is executed.

```

int z = 5;                // L1
while(z < 9)              // L2
{
    if (z > 5)            // L3
    {
        System.out.println(z); // L4
    }
    z = z + 1;           // L5
}

```

Line	z	z < 9 ?	z > 5 ?	print
L1	5			
L2		5 < 9 ? true		
L3			5 > 5 ? false	
L5	6			
L2		6 < 9 ? true		
L3			6 > 5 ? true	
L4				6
L5	7			
L2		7 < 9 ? true		
L3			9 > 5 ? true	
L4				7
L5	8			
L2		8 < 9 ? true		
L3			8 > 5 ? true	
L4				8
L5	9			
L2		9 < 9 ? false		

CORRECT ANSWER:

6
7
8

EXPLANATION:

`z` is 5 and each time round the loop it is increased (note the line `z = z+1;` is in the loop but not in the if statement so it happens every time). The print statement this time is inside the loop so will get printed repeatedly, but only when the test of the if statement lets it. The test of the if statement is only `true` when `z` is more than 5 so the first time the print statement is executed is when `z` has got to 6. The if test is then `true` every time as `z` is increasing so stays greater than 5. However when `z` gets to 9 the loop test becomes `false` so the loop is not entered again.